

FANR 3800: Week 2 Student Notes

Spatial Analysis for Natural Resources · Fall 2026

Invalid Date

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1 Week 2 - Telling the Spatial Story

Name: _____ Date: _____

1.1 How to Use This Sheet: Two-Pass Notes

In class, capture the key idea under each prompt in your own words: a phrase or a sentence, not everything. That evening, reread your notes and use the “After class: add detail” box at

the bottom of each page to add detail from the assigned ESRI Training exercise and the eLC links. Need more room? Use your own paper.

1.2 1 · This Week (10 min)

Weekly schedule, ESRI Training, deliverables, and due dates.

1.3 2 · Lab Review: Last Week's Setup (10 min)

Anything still not working with your ESRI Training or ArcGIS Pro logins?

Why does having professional GIS software matter for someone in my discipline?

1.4 3 · The Week Ahead (5 min)

New ideas this week: *Symbology* (how we show features) and *Map Layout*.

1.5 4 · Application Discussion (10 min) (may vary; instructor's call)

Read the prompt for **your** focus and jot a response; be ready to share.

- **Forestry/Forest Management:** A forester maps timber stand types for a landowner. How does the choice of *Symbology* (colors, patterns) affect a non-GIS-savvy landowner's understanding?
- **Wildlife Management:** Mapping critical red-cockaded woodpecker habitat in a national forest, which essential map elements would you prioritize so managers can interpret it quickly?
- **Fisheries Management:** Mapping fish sampling locations in a reservoir, how can map design distinguish species caught or water-quality values for a non-technical audience?
- **Parks Recreation & Tourism Management:** How would a visitor map's purpose (guidance) change your design choices compared with an internal maintenance map?

1.6 5 · Lecture: *Symbology*, Making Attributes Visible (15 min)

- Symbology is *analysis*, not decoration: the same data with different symbols can tell different stories.

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- The main methods: *Single Symbol*; *Categorized / Unique Values* (qualitative, e.g., species, ownership); *Graduated* (quantitative, e.g., age, basal area) and how class breaks change the message.

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- Choosing well: color for categories vs. sequences; how many classes.

Key terms: *Symbology* · *Categorized* · *Graduated* · *Unique Values* · class breaks

1.7 After Class: Add Detail

From the assigned ESRI Training exercise and the eLC links, add which symbology method best fits a dataset in *your* field, and why.

2 WEDNESDAY

2.1 1 · Lecture: *Map Layout*, the Anatomy of a Credible Map (20 min)

- The essential elements (title, legend, scale bar, north arrow, and data source plus coordinate-system metadata), and what a *missing* element costs the reader's trust.
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- Composition and visual hierarchy: guiding the eye; a map for a report page vs. a map for a wall.
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- how framing, extent, and class breaks can distort; cite sources; show uncertainty.
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- Export and delivery: EXPORT LAYOUT; format and resolution for print vs. screen; consistent file naming.
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Key terms: legend · scale bar · north arrow · metadata · visual hierarchy · EXPORT LAYOUT

2.2 2 · Pre-lab Review: Why This Week's Lab Matters (10 min)

This week's lab has you build and symbolize a map. Why is a clear, credible map a professional deliverable in my field?

2.3 3 · Pre-lab Review: Methods & Tools (10 min)

The tools and steps I will use this afternoon: the SYMBOLOGY pane, the LAYOUT view, and EXPORT LAYOUT.

2.4 4 · Week Summary (10 min)

In your own words, the one thing to remember from Week 2:

Questions I still have (bring to Monday):

Deliverables: ESRI Training certificate due Wed 8:40 AM; lab document due Sunday 11:59 PM.

2.5 After Class: Add Detail

From the assigned ESRI Training exercise and the eLC links, write the checklist of elements you will put on *every* map, and add the one you are most likely to forget.

2.6 Evening Review Checklist (Second Pass)

- ☐ Reread my in-class notes within 24 hours
- ☐ Added detail from the assigned ESRI Training exercise
- ☐ Added detail from the eLC web links
- ☐ Wrote my own one-sentence summary of the week
- ☐ Listed questions to ask on Monday